

SC1007 Data Structures and Algorithms

Lab 6: Searching

Practice Questions

Q1 Given an array of n elements, find two elements in the array such that their sum is equal to K . The two elements can be the same element. Once a pair of elements is found, the program can be terminated. The function prototype is given below:

```
def dual_search(A, size, K, dual_index):  
    #A: The input array of integers.  
    #size: The size of the array.  
    #K: The target sum.  
    #dual_index: A list to store the indices of the two elements.  
    #return true if pairs are found, otherwise return false
```

An example input is:

$A = [3, 1, 7, 4, 5, 9]$

$K = 8$

The output is:

Pair found at indices: $[0, 4]$

Elements: $3 + 5 = 8$

Q2 Given a **sorted** array of n elements (you can use merge sort (Lab 5) to sort the array), find two elements in the array such that their sum is equal to K . The two elements can be the same element. Once a pair of elements are found, the program can be terminated. The results may be different from the results of Question 1. Your algorithm for searching the pair in the sorted array must run in $O(n)$ time complexity. The function prototype is given below:

```
def dual_sorted_search(A, size, K, dual_index):  
    #A: The input array of integers.  
    #size: The size of the array.  
    #K: The target sum.  
    #dual_index: A list to store the indices of the two elements.  
    #return true if pairs are found, otherwise return false
```

An example input is:

A = [9, 1, 4, 3, 7, 5]

K = 8

The output is:

Pair found at indices: [0, 4]

Elements: $1 + 7 = 8$

Q3 Compare the performance between Q1 and Q2. Try to use a large input file to evaluate their running time. A 500k data and 1 million data files are attached. The first line is a search key and the second line is a data size. The rest are the data.